CAPSTONE 15

AI Antenna and RF Concepts Learning Assistant

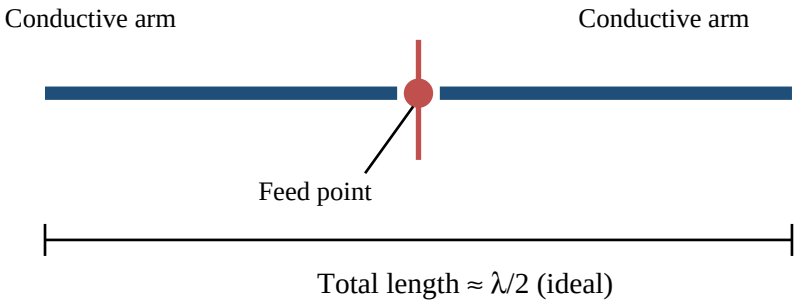
RF Concept Diagrams

Educational diagrams for dipole antennas, radiation patterns, impedance matching, VSWR, and RF link budgets.

Note: These are simplified educational illustrations, not manufacturing drawings or site-specific RF predictions.

1. Dipole antenna

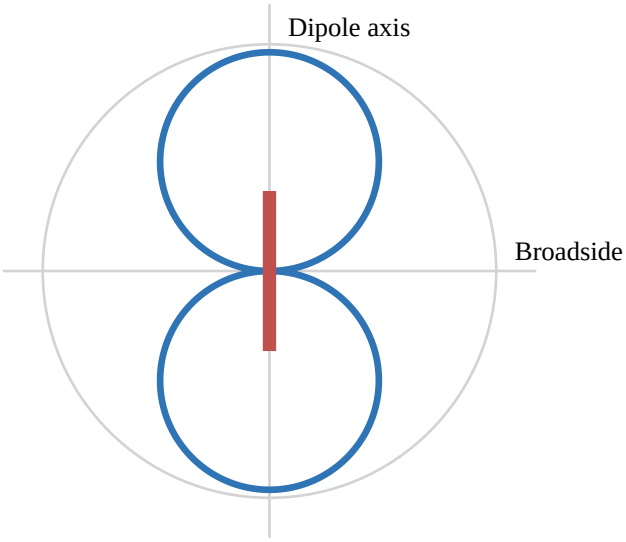
Simplified half-wave dipole (not to scale)



A centre-fed dipole has two conductive arms. An ideal half-wave dipole has a total length near half the free-space wavelength; practical resonant length depends on conductor geometry and surroundings.

2. Idealized dipole radiation pattern

A conceptual 2D cross-section. The dipole axis is vertical; radiation is strongest broadside to the antenna and has nulls along the axis.



Conceptual normalized pattern

The ideal dipole pattern is doughnut-shaped in 3D. This 2D schematic shows a broadside maximum and axial nulls; it is not a measured pattern or a gain scale.

3. Impedance matching concept

A simplified signal chain showing a source, matching network, and load.

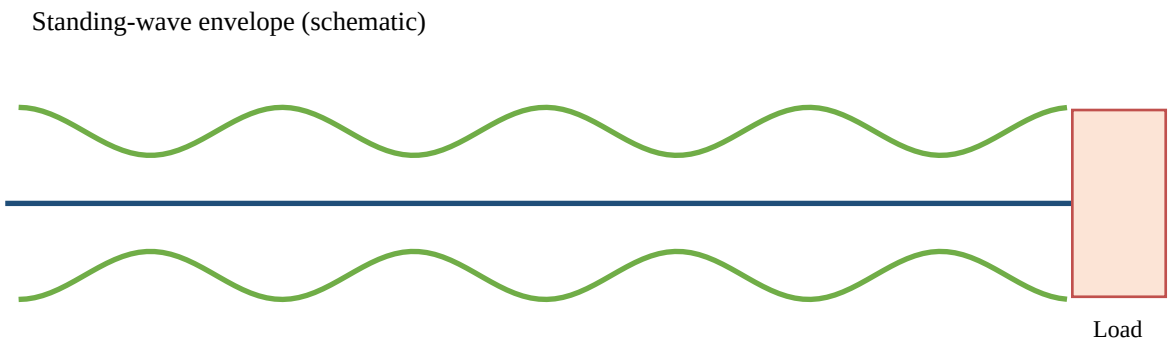
Network transforms impedance

Goal: reduce reflections at the chosen reference plane

A matching network transforms the impedance presented by the load so it is better matched to the source or transmission line at the operating frequency. The correct design depends on frequency and complex impedance.

4. VSWR and reflected waves

Conceptual transmission-line diagram. The standing-wave envelope results from incident and reflected waves.

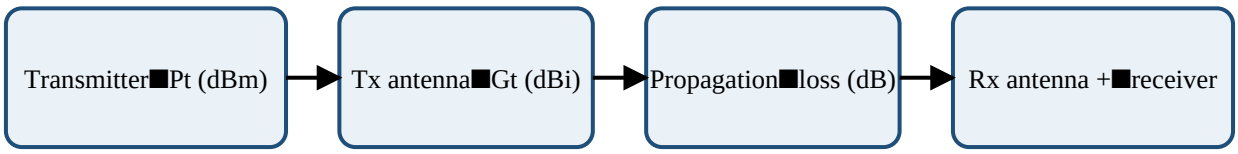


VSWR = Vmax / Vmin; a perfect match gives VSWR = 1

VSWR is the ratio of maximum to minimum voltage magnitude along a transmission line. For a passive load, VSWR is at least 1; a value of 1 indicates no reflected wave in the ideal case.

5. RF link-budget diagram

A simplified path from transmitter to receiver with gains and losses accounted for in dB.



Received power depends on all gains and losses

$$Pr = Pt + Gt + Gr - \text{path loss} - \text{other losses}$$

For a basic link budget, add transmitter power and antenna gains in logarithmic units, then subtract path loss and other applicable losses. Keep units consistent and avoid double-counting losses.